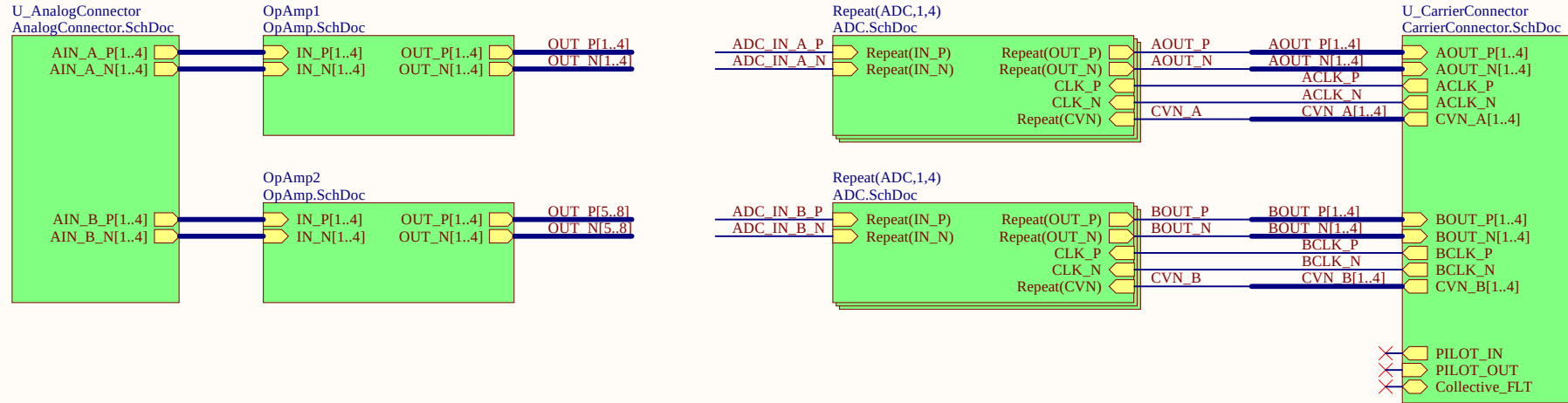




OUT_P1	ADC_IN_A_P1
OUT_N1	ADC_IN_A_N1
OUT_P2	ADC_IN_A_P2
OUT_N2	ADC_IN_A_N2
OUT_P3	ADC_IN_A_P3
OUT_N3	ADC_IN_A_N3
OUT_P4	ADC_IN_A_P4
OUT_N4	ADC_IN_A_N4

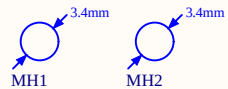
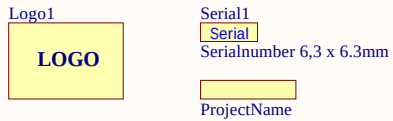
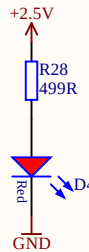
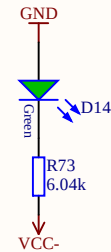
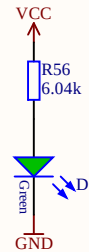
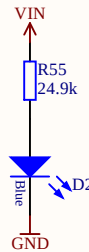
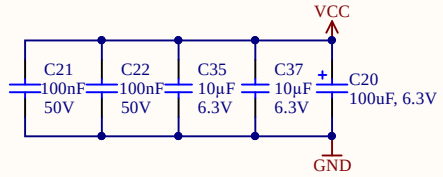
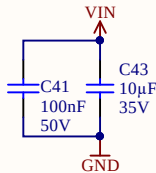
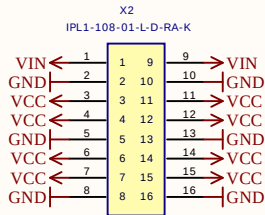
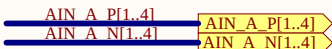
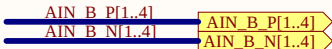
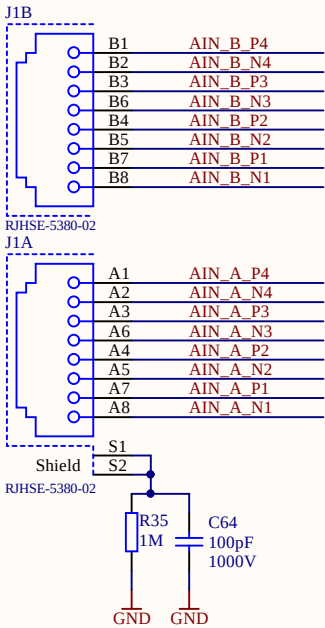
OUT_P5	ADC_IN_B_P1
OUT_N5	ADC_IN_B_N1
OUT_P6	ADC_IN_B_P2
OUT_N6	ADC_IN_B_N2
OUT_P7	ADC_IN_B_P3
OUT_N7	ADC_IN_B_N3
OUT_P8	ADC_IN_B_P4
OUT_N8	ADC_IN_B_N4

Power
Power.SchDoc

A

A

When using the VIN on the power connector X1, use pin 2 and 10 as return ground



Title AnalogConnector.SchDoc		TU München Arcisstraße 21 80333 München GERMANY	
Size: A4	Revision: 3v03		
Date: 02.09.2020	Time: 14:28:14	Sheet 2 of 14	Author: Simon Lukas
Project: UltraZohm_ADC.PrjPCB			

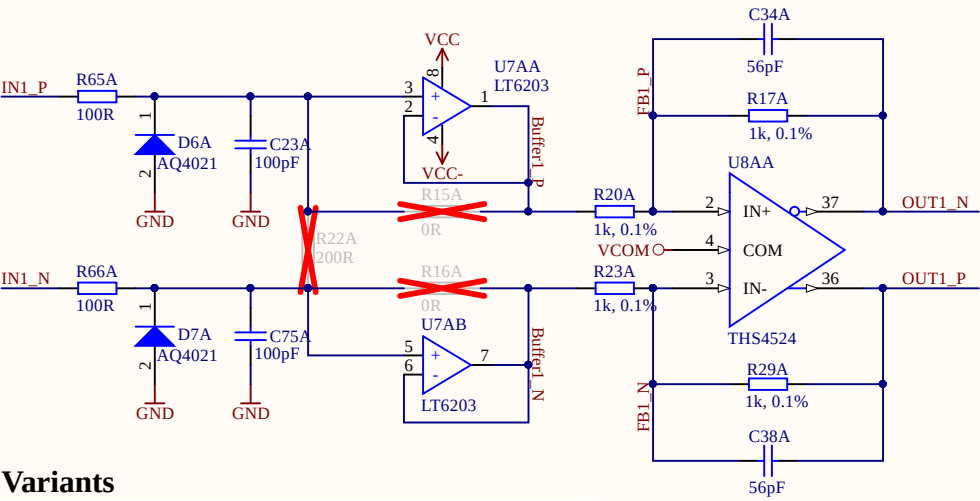


D

D

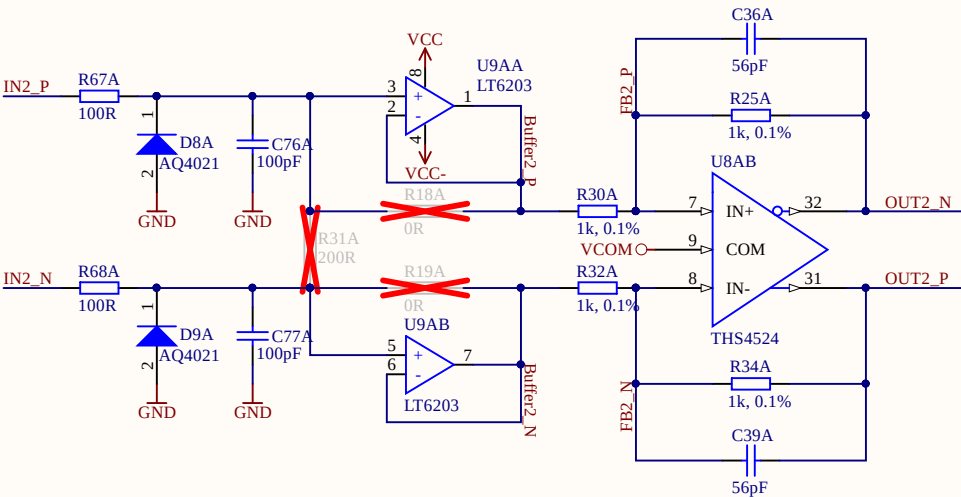
! Use at least 0.1% precision resistors for R20, R23, R17, R29

! Use TVS-diodes GBL05CI in bidirectional configuration

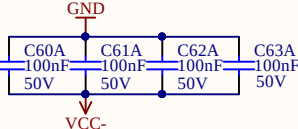
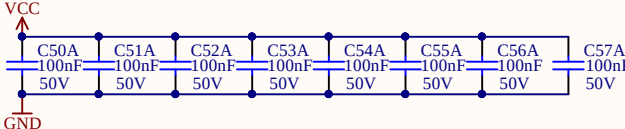
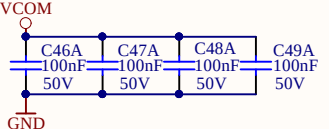
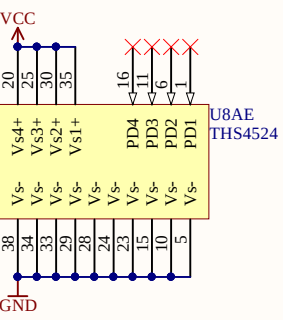
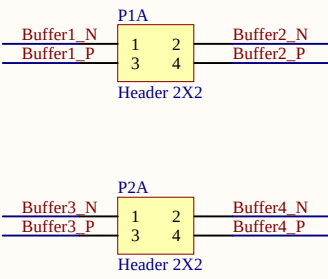
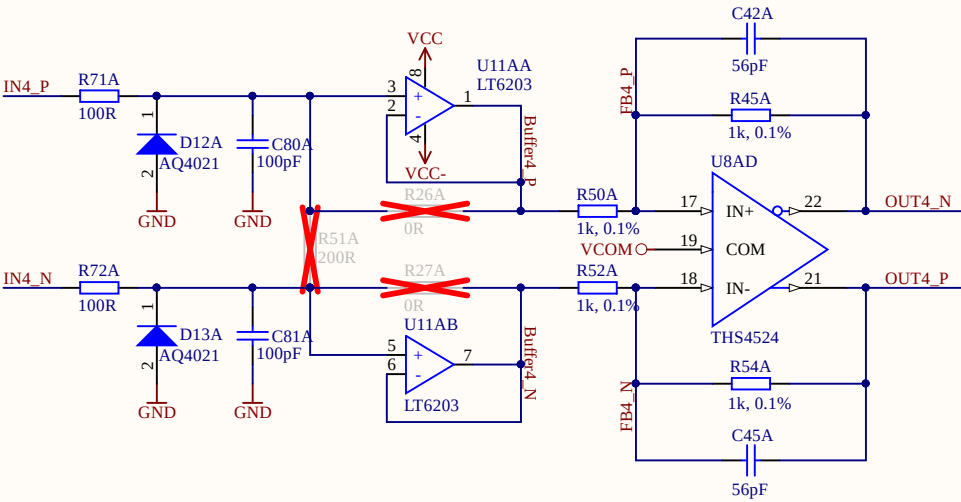
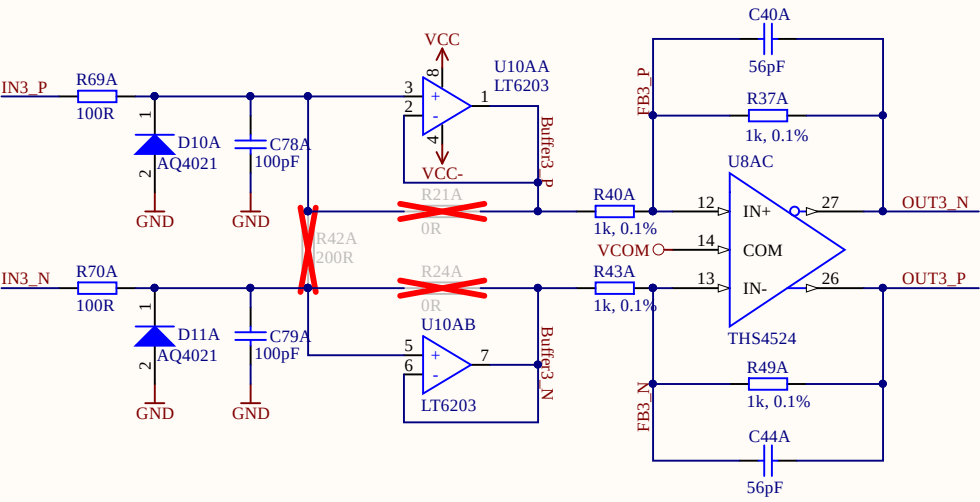


Variants

Without buffer:	R15,R16 = 0R; R65,R66 = 100R;	U7 = dnp R22 = dnp
With buffer:	R15,R16 = dnp; R65,R66 = 100R;	U7 = placed R22 = dnp
Current signal:	R15,R16 = dnp; R65,R66 = 0R;	U7 = placed R22 measurement shunt

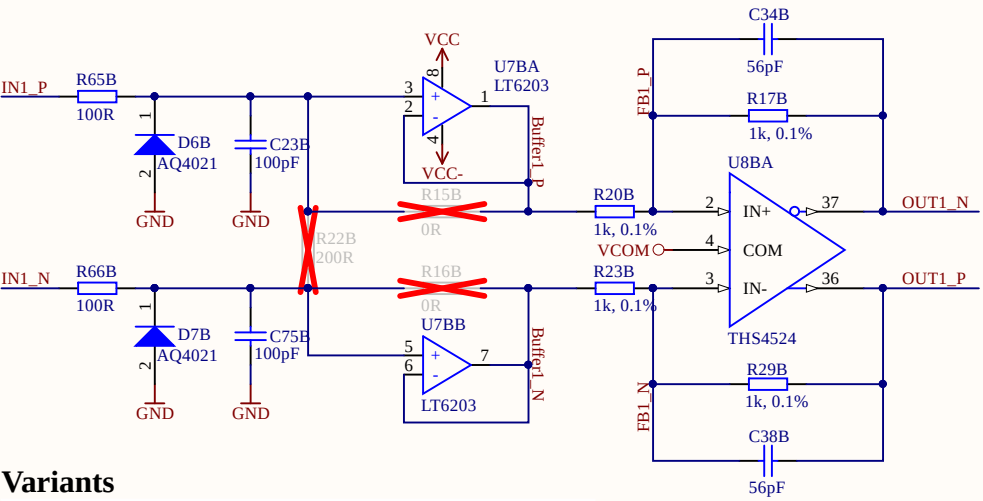


IN_P[1..4]	IN_P[1..4]
IN_N[1..4]	IN_N[1..4]
OUT_P[1..4]	OUT_P[1..4]
OUT_N[1..4]	OUT_N[1..4]
IN1_P	IN_P1
IN1_N	IN_N1
IN2_P	IN_P2
IN2_N	IN_N2
IN3_P	IN_P3
IN3_N	IN_N3
IN4_P	IN_P4
IN4_N	IN_N4
OUT1_P	OUT_P1
OUT1_N	OUT_N1
OUT2_P	OUT_P2
OUT2_N	OUT_N2
OUT3_P	OUT_P3
OUT3_N	OUT_N3
OUT4_P	OUT_P4
OUT4_N	OUT_N4
FB1_P	
FB1_N	
FB2_P	
FB2_N	
FB3_P	
FB3_N	
FB4_P	
FB4_N	
Buffer1_P	
Buffer1_N	
Buffer2_P	
Buffer2_N	
Buffer3_P	
Buffer3_N	
Buffer4_P	
Buffer4_N	



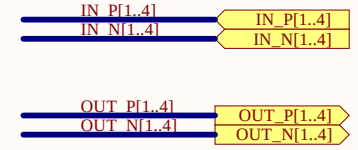
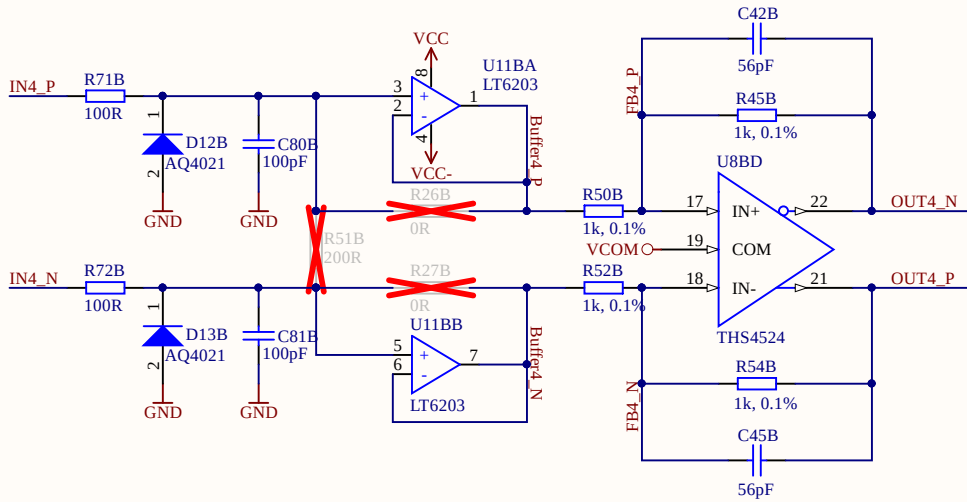
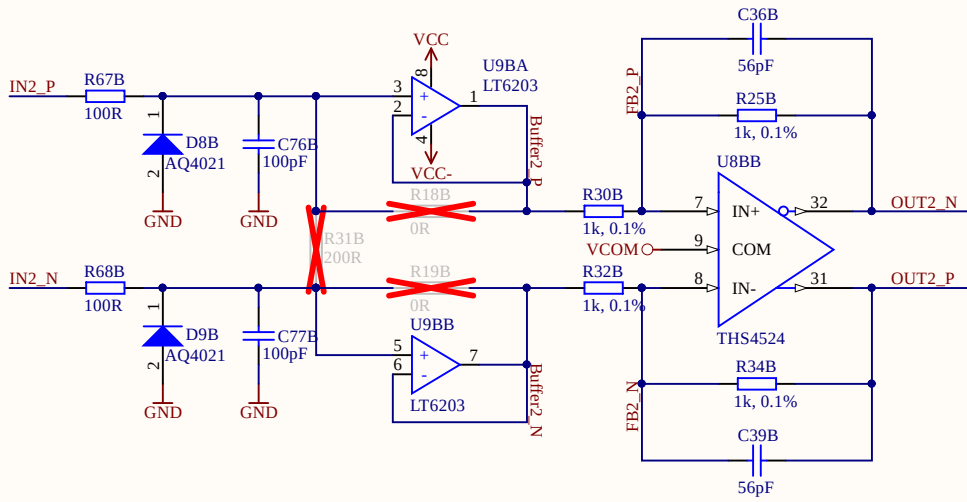
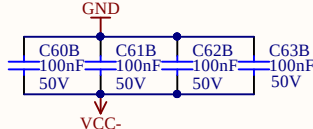
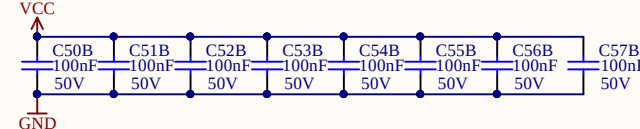
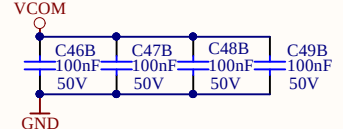
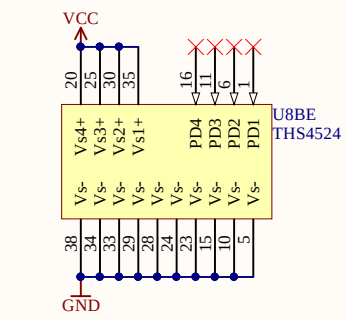
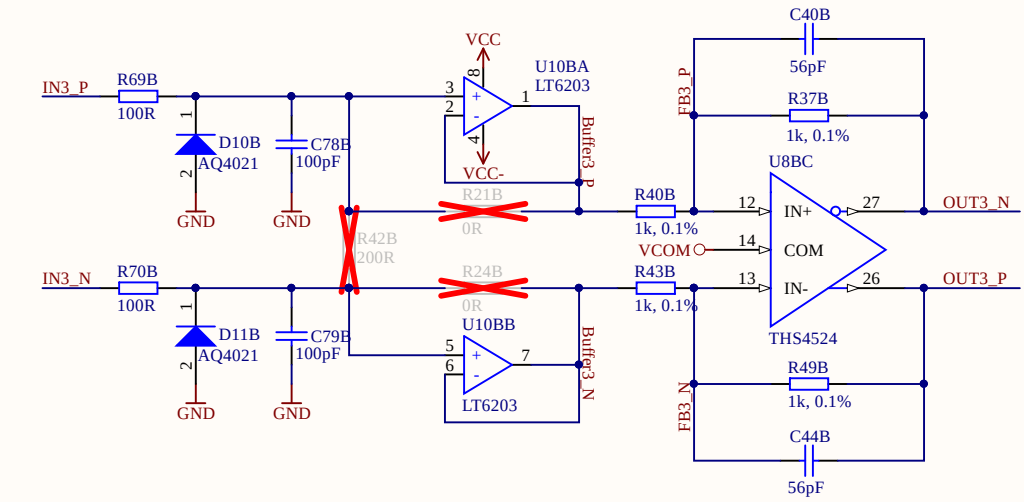
! Use at least 0.1% precision resistors for R20, R23, R17, R29

! Use TVS-diodes GBL05CI in bidirectional configuration

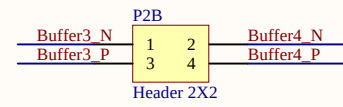
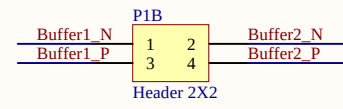


Variants

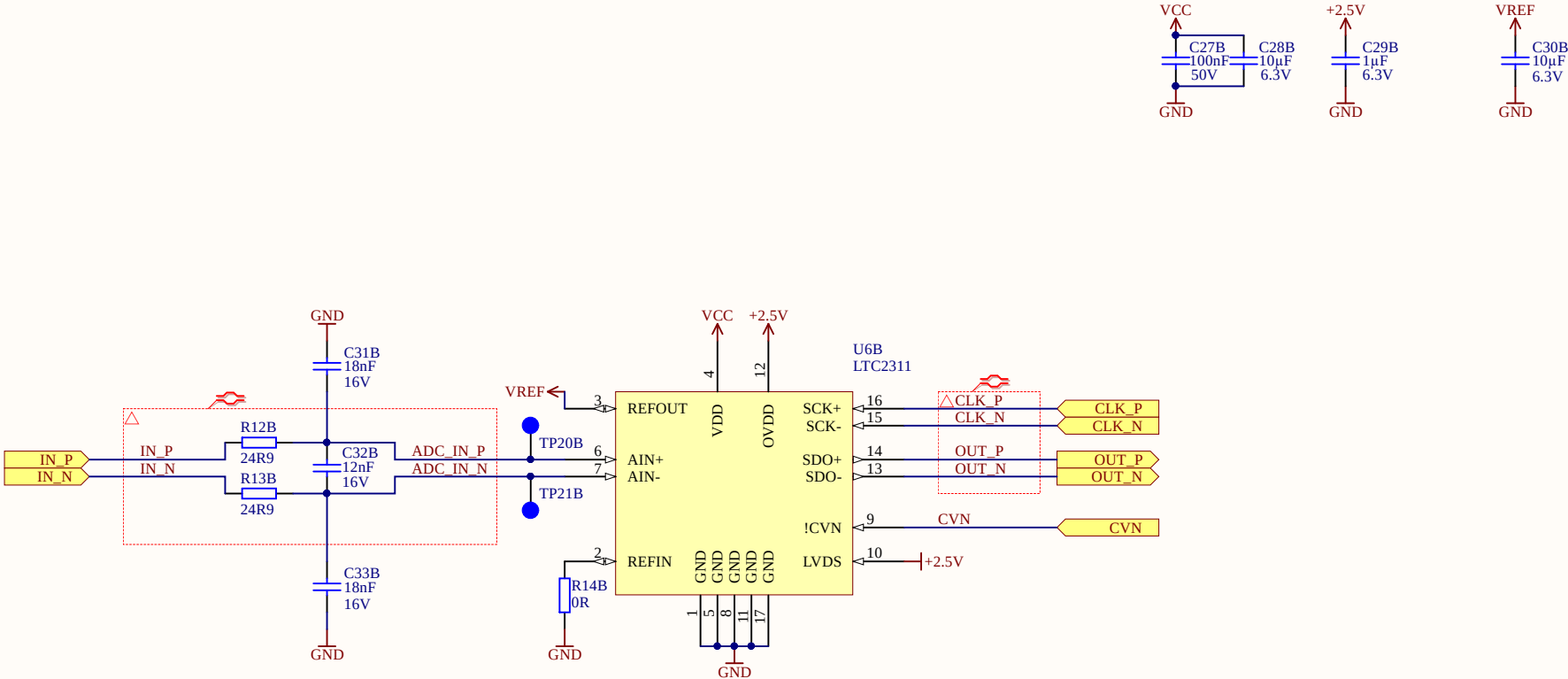
Without buffer:	R15,R16 = 0R; R65,R66 = 100R;	U7 = dnp R22 = dnp
With buffer:	R15,R16 = dnp; R65,R66 = 100R;	U7 = placed R22 = dnp
Current signal:	R15,R16 = dnp; R65,R66 = 0R;	U7 = placed R22 measurement shunt



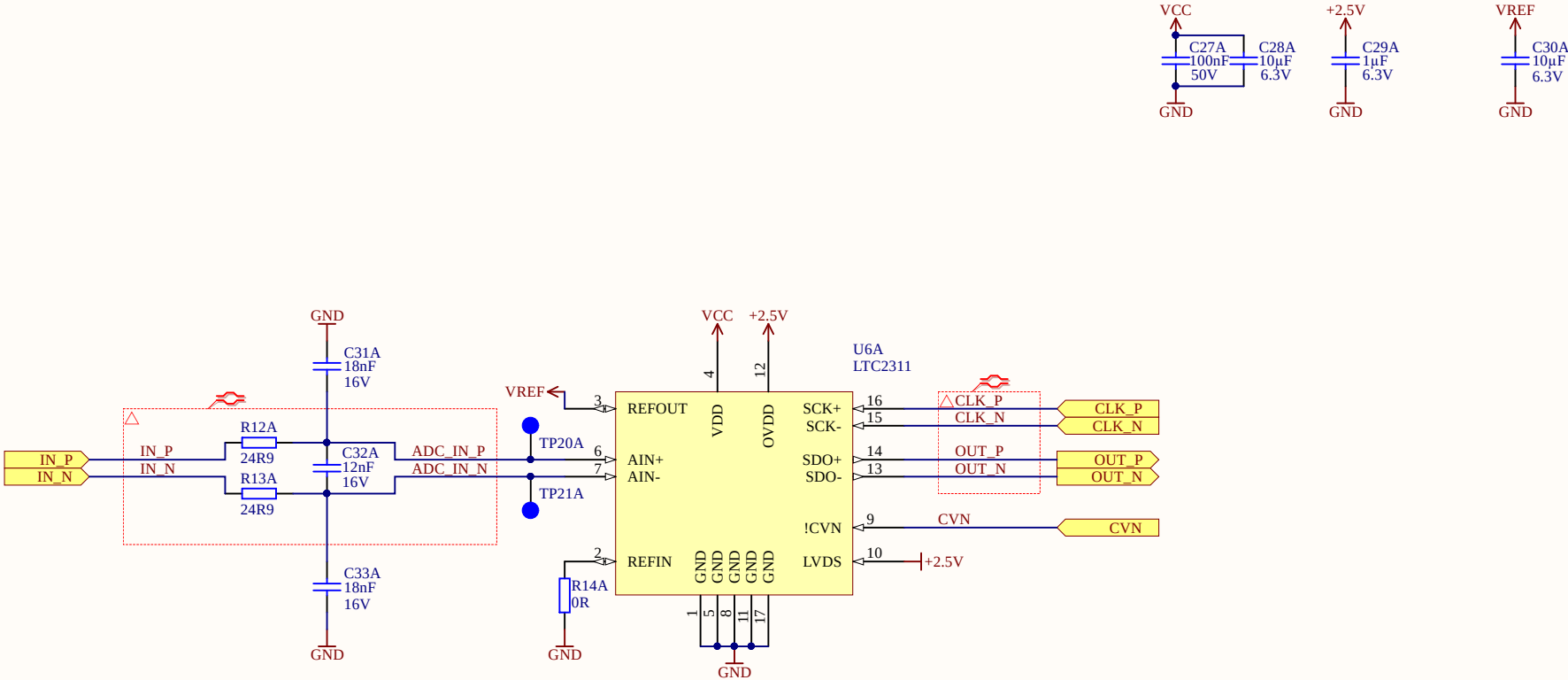
IN1_P	IN_P1
IN1_N	IN_N1
IN2_P	IN_P2
IN2_N	IN_N2
IN3_P	IN_P3
IN3_N	IN_N3
IN4_P	IN_P4
IN4_N	IN_N4
OUT1_P	OUT_P1
OUT1_N	OUT_N1
OUT2_P	OUT_P2
OUT2_N	OUT_N2
OUT3_P	OUT_P3
OUT3_N	OUT_N3
OUT4_P	OUT_P4
OUT4_N	OUT_N4
FB1_P	
FB1_N	
FB2_P	
FB2_N	
FB3_P	
FB3_N	
FB4_P	
FB4_N	
Buffer1_P	
Buffer1_N	
Buffer2_P	
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Buffer3_P	
Buffer3_N	
Buffer4_P	
Buffer4_N	



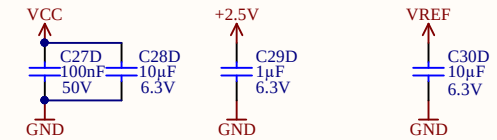
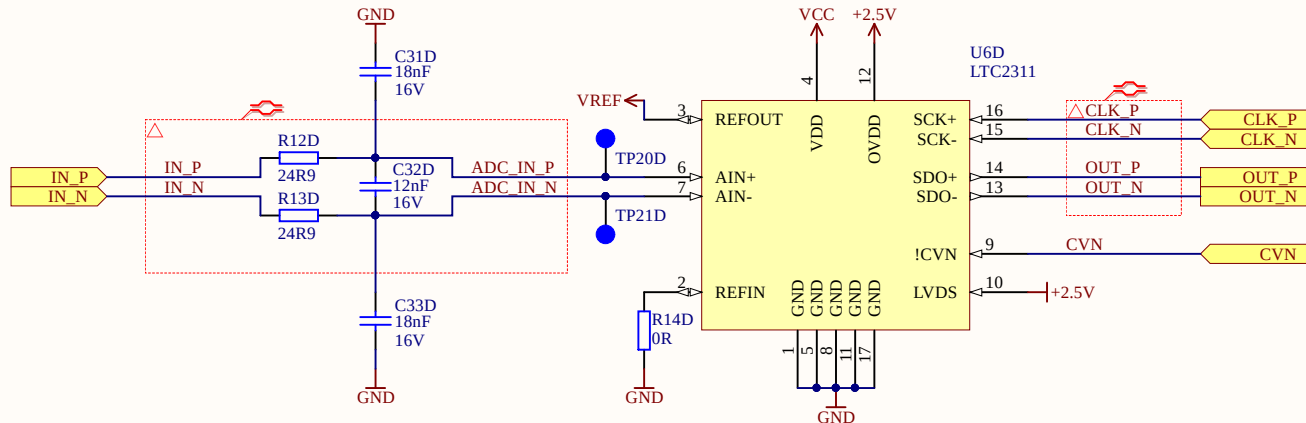
ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



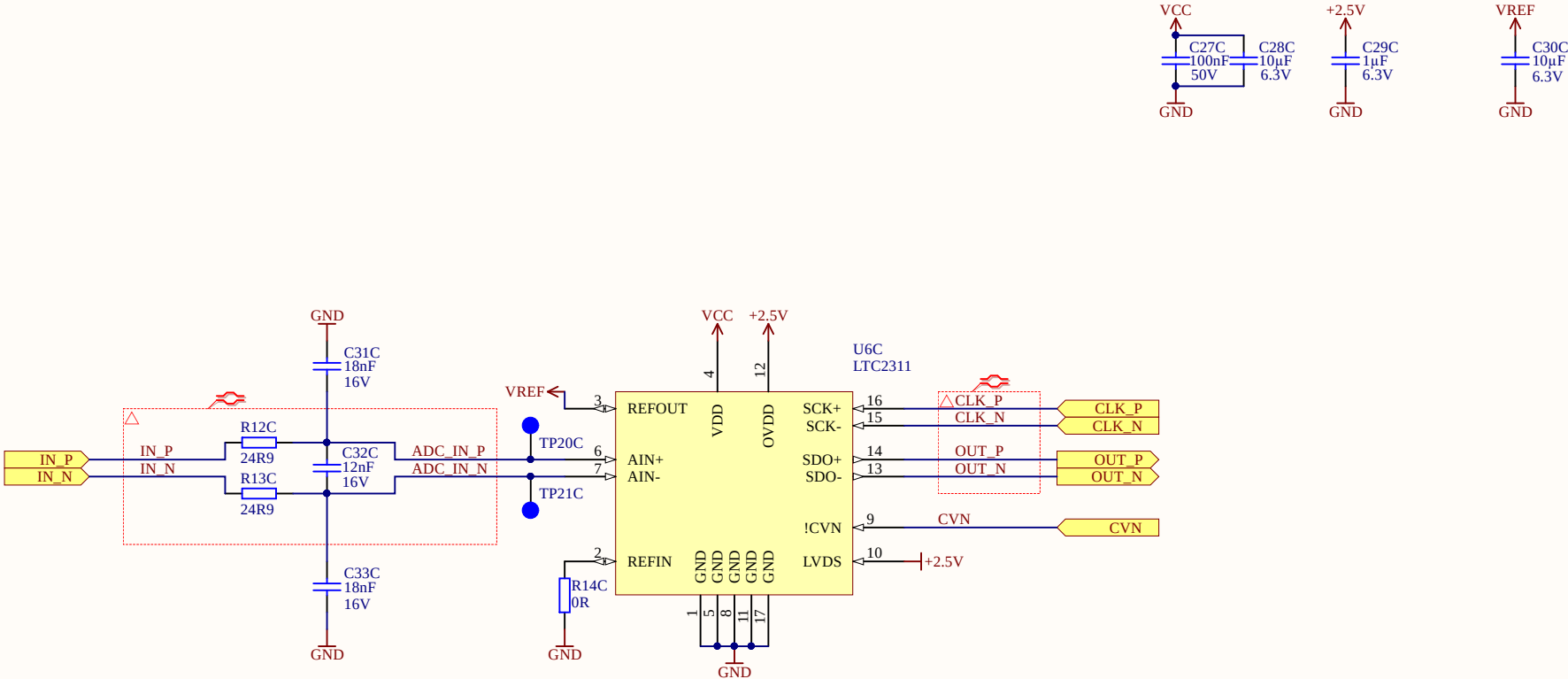
ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



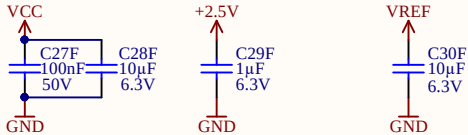
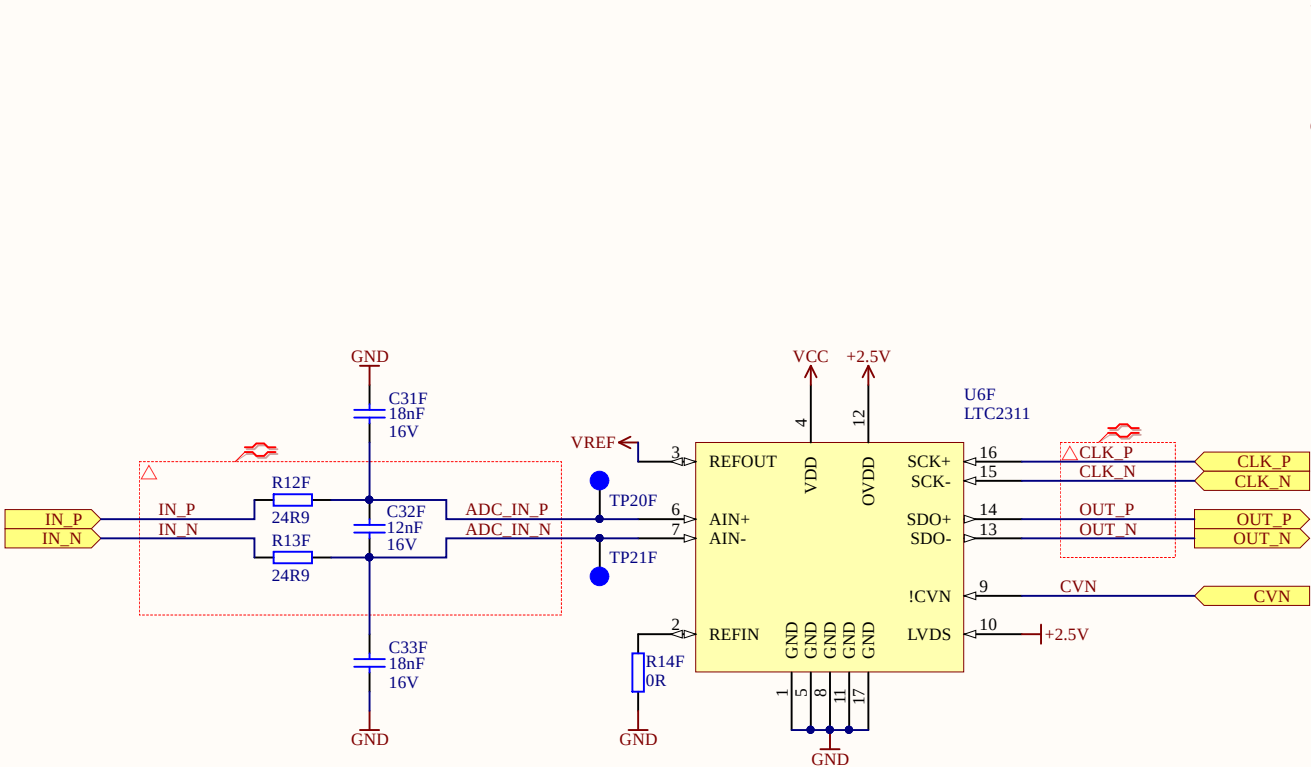
Title: ADC.SchDoc		TU München Arcisstraße 21 80333 München GERMANY		
Size: A4	Revision: 3v03			
Date: 02.09.2020	Time: 14:28:15	Sheet 4.2 of 14	Author: Simon Lukas	
Project: UltraZohm ADC.PriPCB				



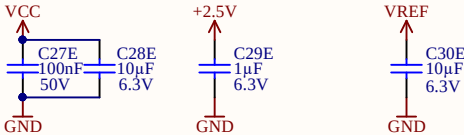
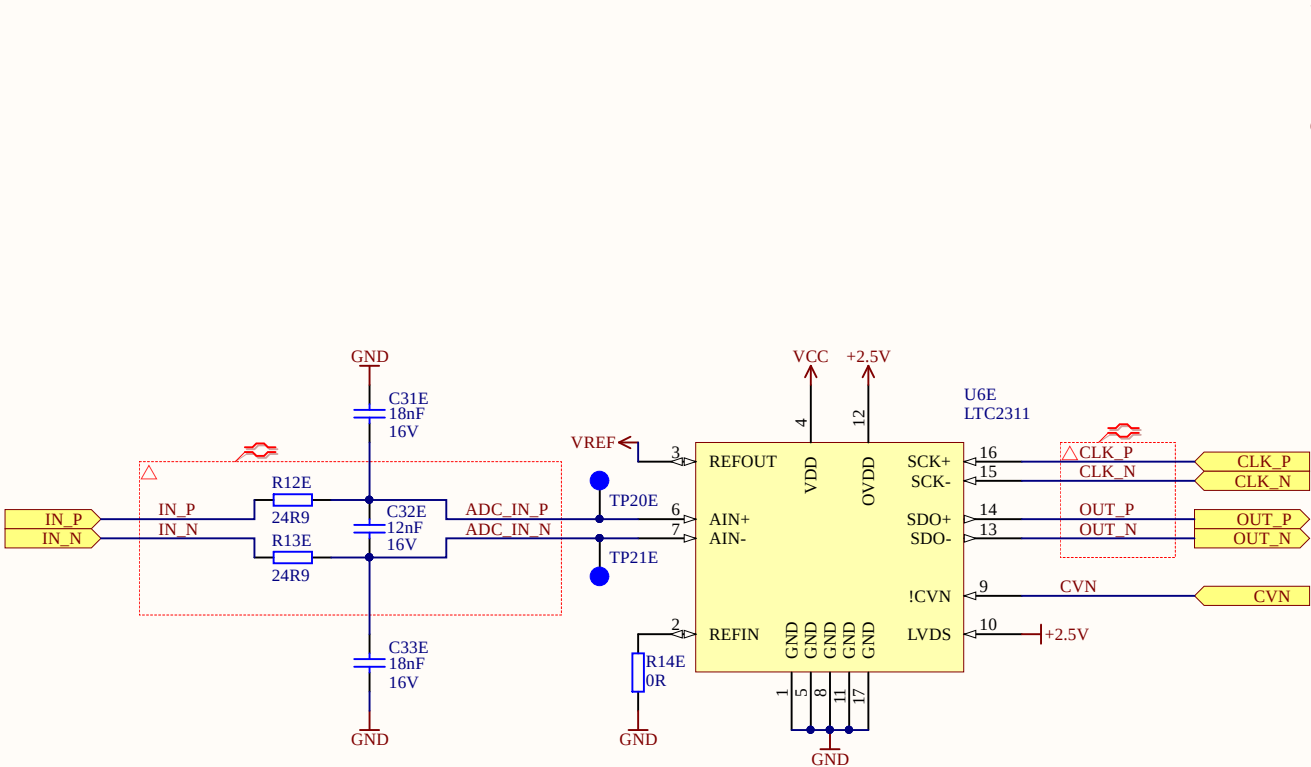
ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



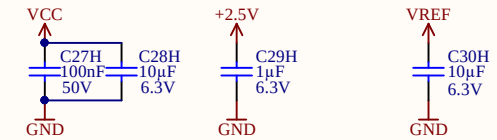
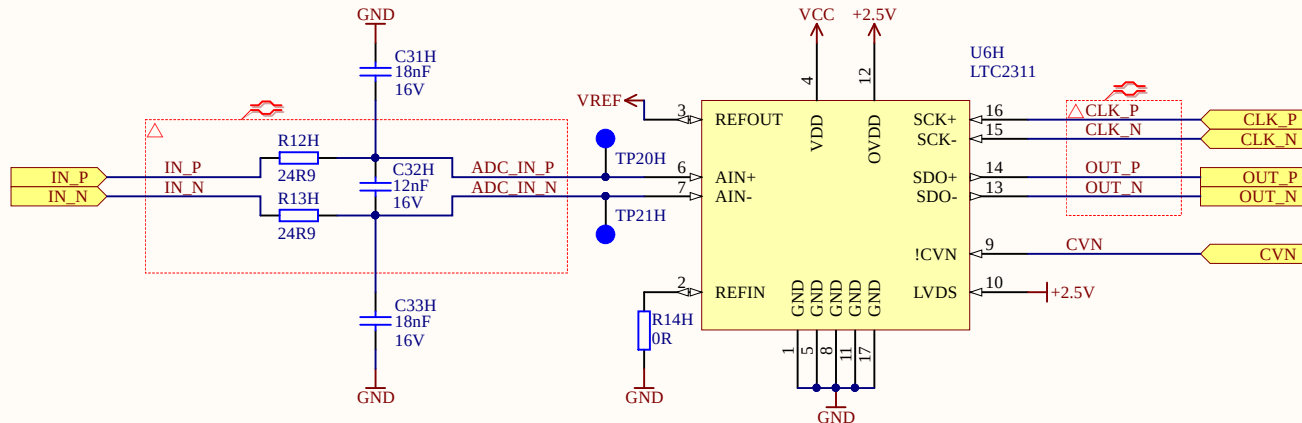
ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



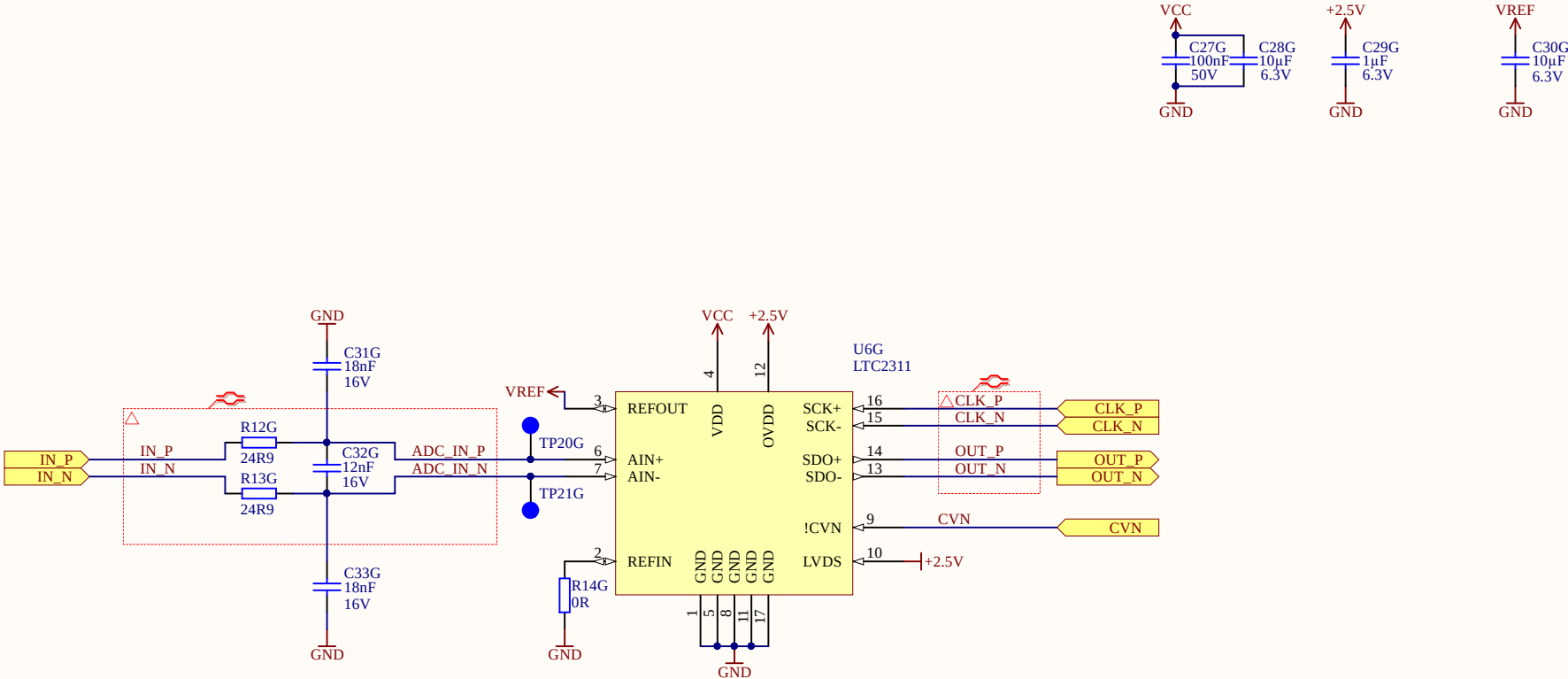
ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



Title: ADC.SchDoc		TU München Arcisstraße 21 80333 München GERMANY		
Size: A4	Revision: 3v03			
Date: 02.09.2020	Time: 14:28:15	Sheet 4.4 of 14	Author: Simon Lukas	
Project: UltraZohm_ADC.PrjPCB				



ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

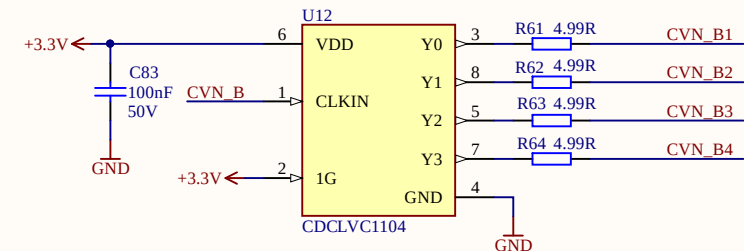
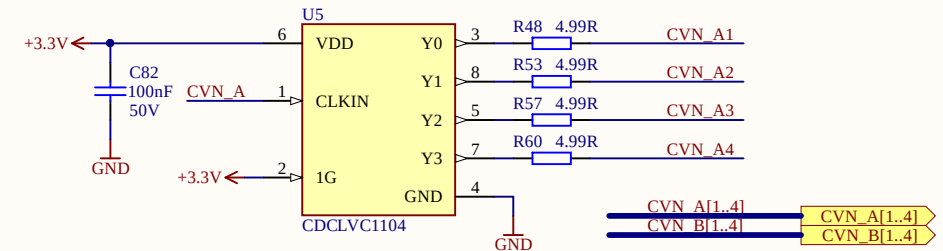
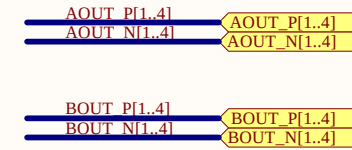
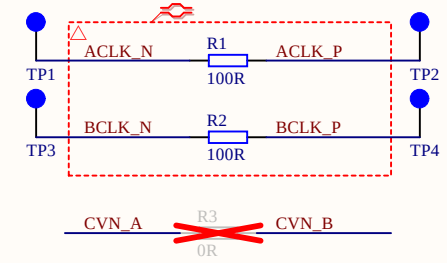
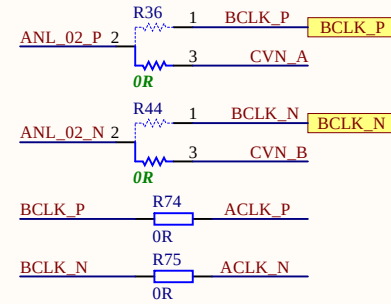
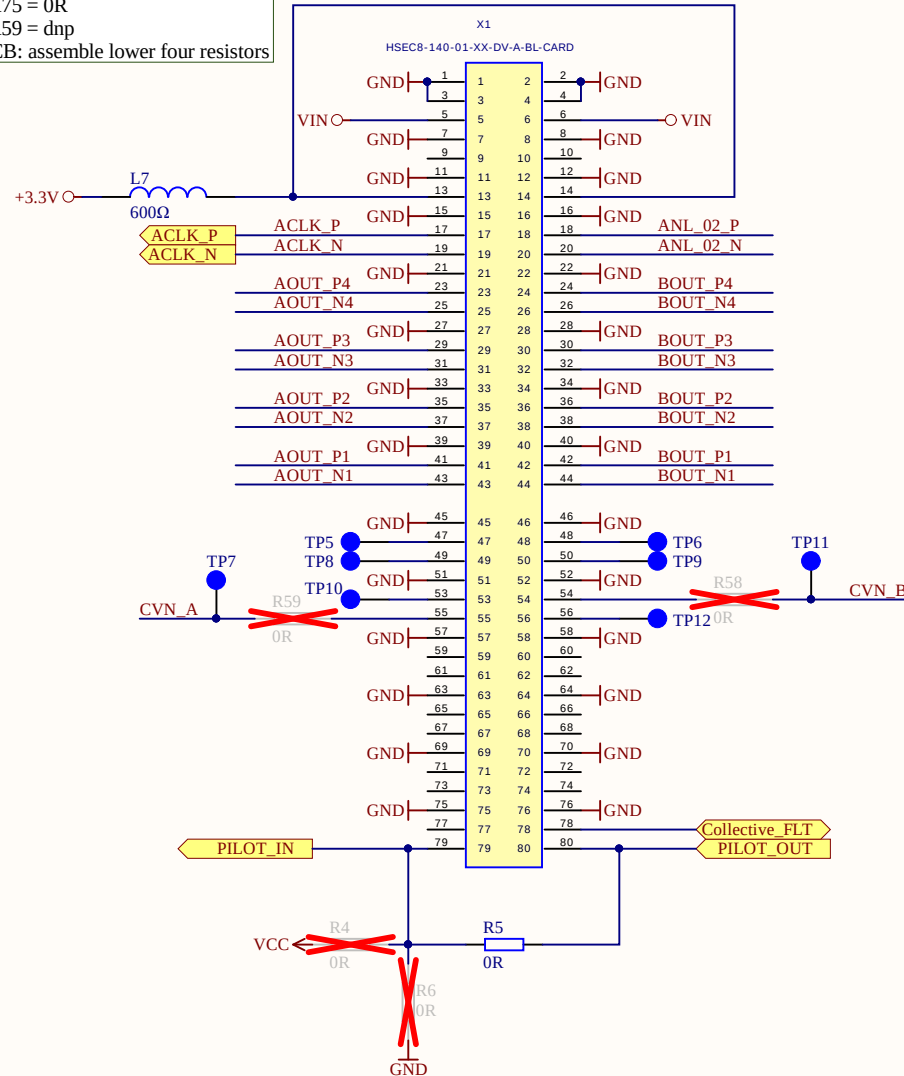


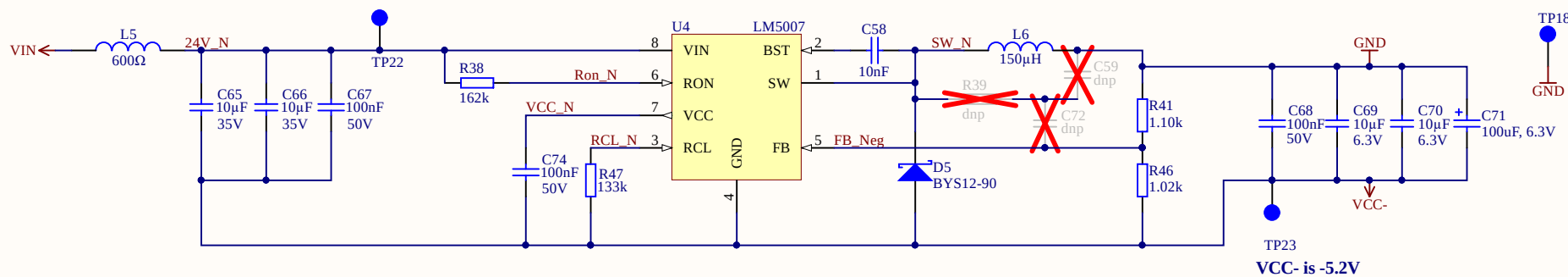
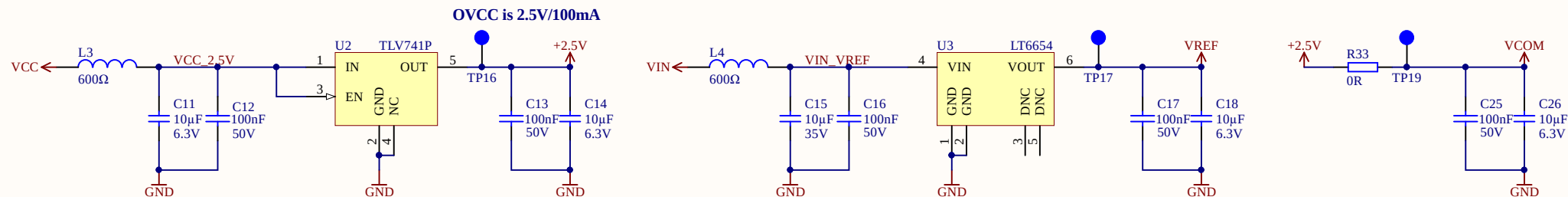
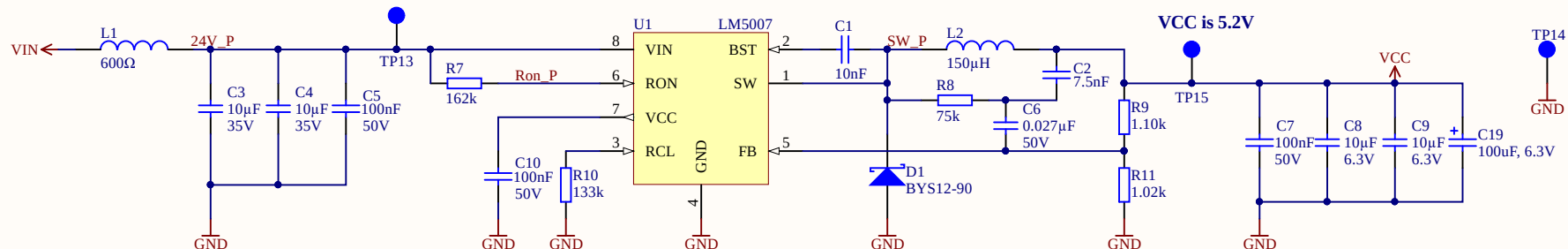
Shared Conversion

```
R36 = up  ANL_02_P -- BCLK_P
R44 = up  ANL_02_N -- BCLK_N
R74,R75 = dnp
R58,R59 = 0R
On PCB: assemble upper two resistors
```

Single Clock Pair

```
R36 = down ANL_02_P -- CVN_A
R44 = down ANL_02_N -- CVN_B
R74,R75 = 0R
R58,R59 = dnp
On PCB: assemble lower four resistors
```





LT6655 is available in 1.25V/2.048V/2.5V/3V/3.3V/4.096V/5V

If the internal reference from LTC2311 (4.096V) is used, U3 must not be fitted

For +/-5.0V rails use 1kOhm resistors (CRCW06031K00JNEAC) for R6, R8, R12, R14

Title Power.SchDoc		TU München Arcisstraße 21 80333 München GERMANY	
Size: A4	Revision: 3v03		
Date: 02.09.2020	Time: 14:28:16	Sheet 6 of 14	Author: Simon Lukas
Project: UltraZohm_ADC.PrjPCB			

